25900561 **Scheme and Answer Key**

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSEPHY203 - INTRODUCTION TO OPTICS

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse (An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any **10** questions; Each question carries **2** marks

1. What is the main postulate of Planck's quantum theory? [K] / [CO1]

ANS: Energy is not continuous but comes in discrete packets called quanta.
Energy of each quantum is proportional to the frequency of radiation (E = hν).
A body can accept or emit integral multiples of quantum

2. Which optical phenomena can be explained by Newton's corpuscular theory of light? [U] / [CO1]

ANS: Reflection and Refraction. (Give full marks to any one answer)

3. Calculate the phase difference between two rays that differ by a path difference of λ/3 [U] / [CO2]

ANS: Phase difference $= \frac{2\pi}{\lambda} \times \frac{\lambda}{3} = \frac{2\pi}{3}$ radians

4. Define geometrical path length of light. [U] / [CO3]

ANS: The shortest distance between any two points in a medium is called the geometric path length.

5. Define a wavefront. Why is it called a phase front? [A] / [CO3]

ANS: A wavefront is a surface over which the optical wave has a constant phase.
It is called a phase front because every point on it oscillates with the same phase.

6. What is the relation between the speed, frequency, and wavelength of light? [A] / [CO3]

ANS: $v = \nu\lambda$

where v is the speed of light, ν is its frequency, and λ is its wavelength

7. What is a secondary wavefront? [U] / [CO4]

ANS: A secondary wavefront is the envelope of secondary wavelets emitted from every point on a primary wavefront.

8. Write the lens-maker's formula for a lens of refractive index n placed in air. [U] / [CO4]

ANS: $(n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) = \frac{1}{f}$

9. What is meant by a ray-transfer matrix? [U] / [CO4]

ANS: A ray-transfer matrix is a 2 × 2 matrix which shows how the position and angle of a light ray change as it passes through an optical system

10. Distinguish between a thick lens and a thin lens. [U] / [CO4]

ANS: A lens is considered "thin" if its thickness is negligible compared to: (1) The radii of curvature of its two surfaces. (2) The distances of objects and images from the lens.

A thick lens is a physically large lens where thickness between its two spherical surfaces is not negligible compared to the radii of curvature of those surfaces. Essentially, its thickness is significant compared to its focal length.

11. An optical system has a numerical aperture (NA) of 0.25 and uses light of wavelength λ=550 nm. Calculate the depth of focus. [A] / [CO6]

ANS: Depth of focus $\frac{\lambda}{(NA)^2} = 8.8\,\mu m$ **OR** Depth of focus $\frac{2\lambda}{(NA)^2} = 17.6\,\mu m$

12. Write down the expression for f-number. [U] / [CO6]

ANS: $f - number = \frac{focal\ length}{diameter\ of\ the\ aperture}$

13. Define the resolving power of a telescope. [U] / [CO6]

ANS: Resolving power is **minimum angular separation** $\theta_{\min}$ between two point-sources that the telescope can distinguish. $\theta_{min} = 1.22\frac{\lambda}{D}$

14. What is the purpose of an aperture in a camera? [U] / [CO6]

ANS: The aperture controls the amount of light entering the camera lens and also affects the depth of field.

[2×10 = 20]

Part B

Short Essay Type Questions

Answer any **6** questions; Each question carries **5** marks

15. Explain Maxwell's electromagnetic theory of light and state its key concepts. [U] / [CO1]

ANS: From the laws of electricity and magnetism, Maxwell predicted the existence of electromagnetic waves. Changing electric and magnetic fields are associated with a light wave. The changing magnetic field produces a time and space varying electric field, and the changing electric field produces a time and space varying magnetic field, and this results in the propagation of the electromagnetic wave even in vacuum.

Light is an electromagnetic disturbance in the form of waves propagated through the aether medium. He estimated the velocity of EM wave (in free space) as $v = \frac{\omega}{k} = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} \approx 3 \times 10^8\ {}^{m}/_{s}$

Key concept: 2 marks, Explanation: 2 marks, Estimation of velocity: 1 mark

16. Light travels from glass (n=1.5) into water (n=1.33). If the wavelength of light in glass is 400 nm, determine: (a) the relative refractive index of water with respect to glass, (b) the wavelength of light in water. [A] / [CO3]

ANS: (a) Relative refractive index = $\frac{n_{water}}{n_{glass}} = \frac{1.33}{1.5} = 0.887$

(b) Wavelength of light in the water, = $\lambda_{water} = \lambda_{glass}\frac{n_{glass}}{n_{water}} = 400\frac{1.5}{1.33} = 451.13\,nm$

Each equation: 1 mark each, Substitution with correct answer and unit: 1.5 marks each

17. The critical angle for a medium is 60° when light passes from this medium into water (n = 1.33). Calculate the refractive index of the medium. What is the condition for total internal reflection in this case? [A] / [CO3]

ANS: (a) $n_1 \sin\theta_1 = n_2 \sin\theta_2$

Refractive index, $n_1 = 1.33 \times \frac{sin\ 90}{sin\ 60} = 1.54$

(b) Condition for total internal reflection:

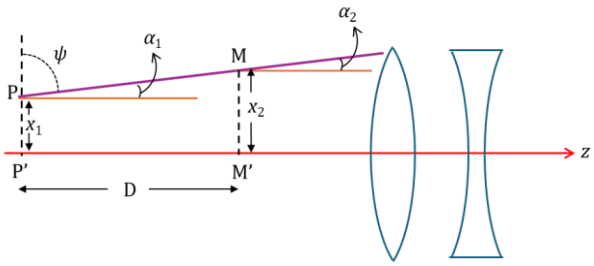
(1) $n_1 > n_2$,

(2) The angle of incidence in the denser medium must be greater than 60°

Equation: 1 mark, Refractive index: 2 marks, Each condition for TIR: 1 mark each

18. Derive the expression for the translation matrix of a paraxial ray traveling through a homogeneous medium. Also, determine the value of the determinant of the translation matrix. [U] / [CO4]

ANS:



$$\begin{bmatrix} \lambda_2 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ D/n_1 & 1 \end{bmatrix} \begin{bmatrix} \lambda_1 \\ x_1 \end{bmatrix}$$

Thus, if a ray is initially specified by a 2×1 matrix with elements λ_1 and x_1, then the effect of translation through a distance D in a homogeneous medium of refractive index n_1, is completely given by the 2×2 matrix :

$$\mathcal{T} = \begin{bmatrix} 1 & 0 \\ D/n_1 & 1 \end{bmatrix}$$

Determinant of the translation matrix is 1.

Derivation of the expression for translation matrix:4 marks, Determinant: 1 mark

19. An object is placed 15 cm in front of a thin lens and forms a real image 30 cm on the other side. What is the focal length of the lens? [A] / [CO4]

ANS: Object distance u = -15cm

Image distance v = 30 cm

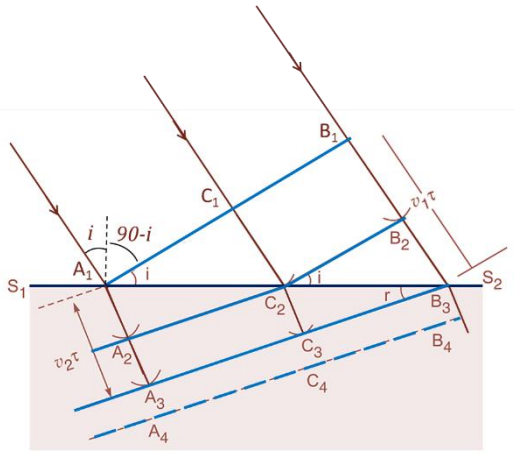
$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

f = 10cm

Equation only 2 marks, Correct answer with unit: 3 marks

20. Derive the law of refraction (Snell's law) of a plane wave using Huygens' principle. [A] / [CO5]

ANS:



$$\frac{\sin i}{\sin r} = \frac{v_1(\tau - \tau_1)}{v_2(\tau - \tau_1)} = \frac{v_1}{v_2}$$

Figure: 1 mark, Derivation: 4 marks

21. Describe chromatic aberration in lenses and how one can reduce it? [U] / [CO6]

ANS: Variation of the image distance from the lens with refractive index measures axial or longitudinal chromatic aberration and the variation in the size of the image measures lateral chromatic aberration.

Definition only, 2 marks + Explanation with figure 3 marks

22. An astronomical telescope has an objective lens of focal length 10 m and an eyepiece of focal length 10 cm. Calculate the telescope's tube length and its magnification. [U] / [CO6]

ANS: Telescope's tube length, or the distance between the objective and the eyepiece, $d = f_o + f_e$ = 10m+10cm = 1010cm

Magnification, $M = \frac{f_o}{f_e}\frac{1000}{10} = 100$

Each equation 1 mark, substitution with correct answer: 1.5 marks each.

[5×6 = 30]